
بسم اللہ الرحمن الرحیم
صلی اللہ علیہ وسلم
Iqbal group of bs zoology
03460004097
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Muhammad Iqbal arain
quiz no
شب ترنم رہے گا اشکوں کے اختصار تک
اقبال
اگر پھر بھی خود کو تنہا پاؤ تو آواز دینا
2022


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ZOO101:Quiz 2

Quiz Start Time: 05:52 PM, 30 June 2022

Question # 1 of 10 (Start time: 05:52:27 PM, 30 June 2022)

Total Marks: 1

Part of an optical system, which is directed to the viewer is_____.

Select the correct option

- (●) Eyepiece
- ( ) Lens tube
- ( ) All of above
- ( ) Diopter

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 05:52 PM, 30 June 2022

Question # 2 of 10 (Start time: 05:53:58 PM, 30 June 2022)

Total Marks: 1

Most common fixative -fixation of biopsy specimen?

Select the correct option

- [x] Formaldehyde
- [ ] Alcohol
- [ ] Glutaraldehyde
- [ ] All of the above

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 05:52 PM, 30 June 2022

Question # 3 of 10 (Start time: 05:54:36 PM, 30 June 2022) Total Marks: 1

Zenker's fixative contains?

Select the correct option

- Acetic acid
- Water
- Potassium dichromate
- All of the above

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 05:52 PM, 30 June 2022

Question # 4 of 10 (Start time: 05:55:49 PM, 30 June 2022) Total Marks: 1

In which of the following microtome, tissue requires little preparation.

Select the correct option

- Sliding microtome
- Cryostat microtome
- Rotary rocking microtome
- Saw microtome

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 05:52 PM, 30 June 2022

Question # 5 of 10 (Start time: 05:56:46 PM, 30 June 2022) Total Marks: 1

Which of the following is fast freezing chemical?

Select the correct option

- Water
- Liquid Nitrogen
- Ether
- Isoamyl alcohol

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Click to Save Answer & Move to Next Question

Time Left sec(s)

ZOO101:Quiz 2

Quiz Start Time: 05:52 PM, 30 June 2022

Question # 6 of 10 (Start time: 05:57:39 PM, 30 June 2022)

Total Marks: 1

The rate of migration of form I DNA than form III DNA under some conditions is

Select the correct option

- [x] Faster
- [ ] Neither fast nor slow
- [ ] Slower
- [ ] Zero

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 05:52 PM, 30 June 2022

Question # 8 of 10 (Start time: 06:00:33 PM, 30 June 2022)

Total Marks: 1

The typical cut thickness for a rotary microtome is between _____.

Select the correct option

- 2 and 20 µm
- 2 and 60 µm
- 3 and 50 µm
- 1 and 60 µm

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Click to Save Answer & Move to Next Question

Time Left sec(s)

ZOO101:Quiz 2

Quiz Start Time: 05:52 PM, 30 June 2022

Question # 9 of 10 (Start time: 06:01:33 PM, 30 June 2022)

Total Marks: 1

The smaller is the wavelength of light (λ)

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Select the correct option

- The smaller is its ability to resolve the points on the object
- The greater is its ability to resolve the points on the object

Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 05:52 PM, 30 June 2022

Question # 10 of 10 (Start time: 06:02:46 PM, 30 June 2022) Total Marks: 1

How many forms of DNAs are migrate through agarose gels at different rates?

Select the correct option

- 2
- 4
- 5
- 3

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Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 06:11:31 PM, 30 June 2022) Total Marks: 1

The vibrating microtome operates by cutting using a vibrating blade, allowing the resultant cut to be made with how much pressure than would be required for a stationary blade

Select the correct option

- Zero pressure
- Less pressure
- High pressure
- Zero pressure

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 06:11 PM, 30 June 2022

Question # 3 of 10 (Start time: 06:14:42 PM, 30 June 2022)

Total Marks: 1

Dark field microscopy is a technique used in _____.

Select the correct option

- None
- Light microscopy
- Light microscopy and Electron microscopy
- Electron microscopy

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 06:11 PM, 30 June 2022

Question # 4 of 10 (Start time: 06:15:34 PM, 30 June 2022)

Total Marks: 1

The practical limit to magnification with a light microscope is around_____.

Select the correct option

- 1900X
- 1300X
- 1400X
- 1700X

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Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 06:16:10 PM, 30 June 2022) Total Marks: 1

Who was the inventor of confocal microscopy?

Select the correct option

- Washington Teasdale
- Marvin Minsky
- None of the above
- Antonie van Leeuwenhoek

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Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 06:16:54 PM, 30 June 2022) Total Marks: 1

In coagulative fixative, removal of water destabilizes

Select the correct option

- All of the above
- Hydrogen bonding
- Covalent bonding
- Ionic bonding

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 06:11 PM, 30 June 2022

Question # 7 of 10 (Start time: 06:17:45 PM, 30 June 2022)

Total Marks: 1

Which of the following is used in certain specific histological preparations

Select the correct option

- Chromic acid
- All of the above
- Potassium dichromate
- Potassium permanganate

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 06:11 PM, 30 June 2022

Question # 8 of 10 (Start time: 06:18:29 PM, 30 June 2022) Total Marks: 1

In confocal microscopy, the light returning from the specimen would pass through a _____.

Select the correct option

- First pinhole aperture
- All of the above
- Second pinhole aperture
- Third pinhole aperture

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 06:11 PM, 30 June 2022

Question # 9 of 10 (Start time: 06:19:40 PM, 30 June 2022) Total Marks: 1

Which knives are used to cut ultrathin slices of samples for electron and light microscope?

Select the correct option

- Steel knives
- Glass knives
- Sapphire
- Diamond knives

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 06:11 PM, 30 June 2022

Question # 10 of 10 (Start time: 06:20:41 PM, 30 June 2022)

Total Marks: 1

Part of an optical system, which is directed to the viewer is_____.

Select the correct option

- All of above
- Lens tube
- Diopter
- Eyepiece

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 10:39 PM, 30 June 2022

Question # 1 of 10 (Start time: 10:39:45 PM, 30 June 2022) Total Marks: 1

The part that anchor the knife holder to the microtome stage is called

Select the correct option

- Clamp
- knife holder
- Sapphire
- Knife holder base

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 10:39 PM, 30 June 2022

Question # 2 of 10 (Start time: 10:40:49 PM, 30 June 2022)

Total Marks: 1

An aspheric lens avoids _____.

Select the correct option

- Fine adjustment
- All of above
- Intensity of light
- Chromatic aberration

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 10:39 PM, 30 June 2022

Question # 4 of 10 (Start time: 10:42:41 PM, 30 June 2022) Total Marks: 1

_____ is a suitable method for preserving molecular features of samples.

Select the correct option

- Chemical method
- Fixation by heating
- Quick freezing
- Physical method

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Quiz Start Time: 10:39 PM, 30 June 2022

Question # 5 of 10 (Start time: 10:43:07 PM, 30 June 2022)

Total Marks: 1

SYBR Gold method can detect ds DNA in Agarose gel up to

Select the correct option

- ○ >20pg
- ● <20pg
- ○ 20pg
- ○ 10pg

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Click to Save Answer & Move to Next Question

sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:52 PM, 30 June 2022

Question # 2 of 10 (Start time: 10:54:59 PM, 30 June 2022)

Total Marks: 1

In which year, Schwartz and Cantor developed pulsed-field gel?

Select the correct option

- 1948
- 1989
- 1849
- 1984

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2

Question # 3 of 10 (Start time: 10:55:34 PM, 30 June 2022)

For contact free slicing of tissue section which type of microtome is used?

Select the correct option

- ○ Laser microtome
- ● Vibrating microtome
- ○ Rotory microtome
- ○ Ultramicrotome

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ZOO101:Quiz 2

Quiz Start Time: 10:52 PM, 30 June 2022

Question # 4 of 10 (Start time: 10:57:03 PM, 30 June 2022)

Total Marks: 1

The ocular micrometer is a glass disc with _____ equal divisions or lines on it but with no absolute value.

Select the correct option

- 10
- 100
- 10000
- 1000

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Click to Save Answer & Move to Next Question

ZOO101:Quiz 2 Quiz Start Time: 10:52 PM, 30 June 2022

Question # 5 of 10 (Start time: 10:57:34 PM, 30 June 2022) Total Marks: 1

Occular micrometer is used inside the _____.

Select the correct option

- Base
- Objective lens
- Eyepiece
- Illuminator

Click to Save Answer & Move to Next Question

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ZOO101:Quiz 2

Quiz Start Time: 10:52 PM, 30 June 2022

Question # 8 of 10 (Start time: 11:01:30 PM, 30 June 2022)

Total Marks: 1

Which knives are used to cut ultrathin slices of samples for electron and light microscope?

Select the correct option

- Diamond knives
- Glass knives
- Steel knives
- Sapphire

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Click to Save Answer & Move to Next Question

10:32 4G 29%
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2

BC200203609: MUHAMMAD IQBAL

Time Left 69 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 9 of 10 (Start time: 10:32:22 PM, 30 June 2022)

Total Marks: 1

How much size of DNA fragments are best resolved in Agarose gels?

Select the correct option

- ○ 30-20,000bp
- ● 50-20,000bp
- ○ 20-50,000bp
- ○ 50-50,000bp

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Click to Save Answer & Move to Next Question

Introduction

Agarose Gel

- Small DNA fragments (50-20,000 bp) are best resolved in agarose gels run in a horizontal configuration in an electric field of constant strength and direction.
- As size of DNA fragment increases the percentage of gel composition have to be decreased.

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Introduction

Gel composition

The greater the pore size of the gel, the larger the DNA that can be

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BC200203609: MUHAMMAD IQBAL
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ZOO101:Quiz 2
Quiz Start Time: 10:23 PM, 30 June 2022

Question # 8 of 10 (Start time: 10:31:11 PM, 30 June 2022) Total Marks: 1

To observe the sample through a fluorescence microscope, it should be first labeled with

Select the correct option

- ○ Fluorophore
- ○ Giemsa
- ○ Iodine
- ● All of above

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Click to Save Answer & Move to Next Question

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BC200203609: MUHAMMAD IQBAL

Time Left 45 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 10 of 10 (Start time: 10:32:57 PM, 30 June 2022)

Total Marks: 1

When electrical voltage is applied, the DNA migrate to which direction?

Select the correct option

- ○ Negative anode
- ● Positive anode
- ○ Negative cathode
- ○ Positive cathode

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Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:30:30 PM, 30 June 2022) Total Marks: 1

The great advantage of ethidium bromide is the ability to examine the Agarose gels directly by

Select the correct option

- ○ Imaging
- ● UV illumination
- ○ Microscope
- ○ X-rays radiations

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Click to Save Answer & Move to Next Question

Detection of DNA in Agarose gel

Advantage over Disadvantage

- electrophoretic mobility of linear dsDNA reduced by -15% in the presence of the dye
- the ability to examine the agarose gels directly under UV illumination during or at the end of the run is a great advantage.

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Detection of DNA in Agarose gel

Advantage over Disadvantage

- Sharper DNA bands are

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BC200203609: MUHAMMAD IQBAL

Time Left 40 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 6 of 10 (Start time: 10:29:13 PM, 30 June 2022)

Total Marks: 1

The relation between the effective ranges of separation in Agarose gel to the voltage is

Select the correct option

- ○ Either directly or inversely
- ○ Directly proportional
- ● Neither directly nor inversely
- ○ Inversely proportional

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Saving...

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BC200203609: MUHAMMAD IQBAL

Time Left 71 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 5 of 10 (Start time: 10:28:32 PM, 30 June 2022)

Total Marks: 1

SYBR Gold method has high affinity for?

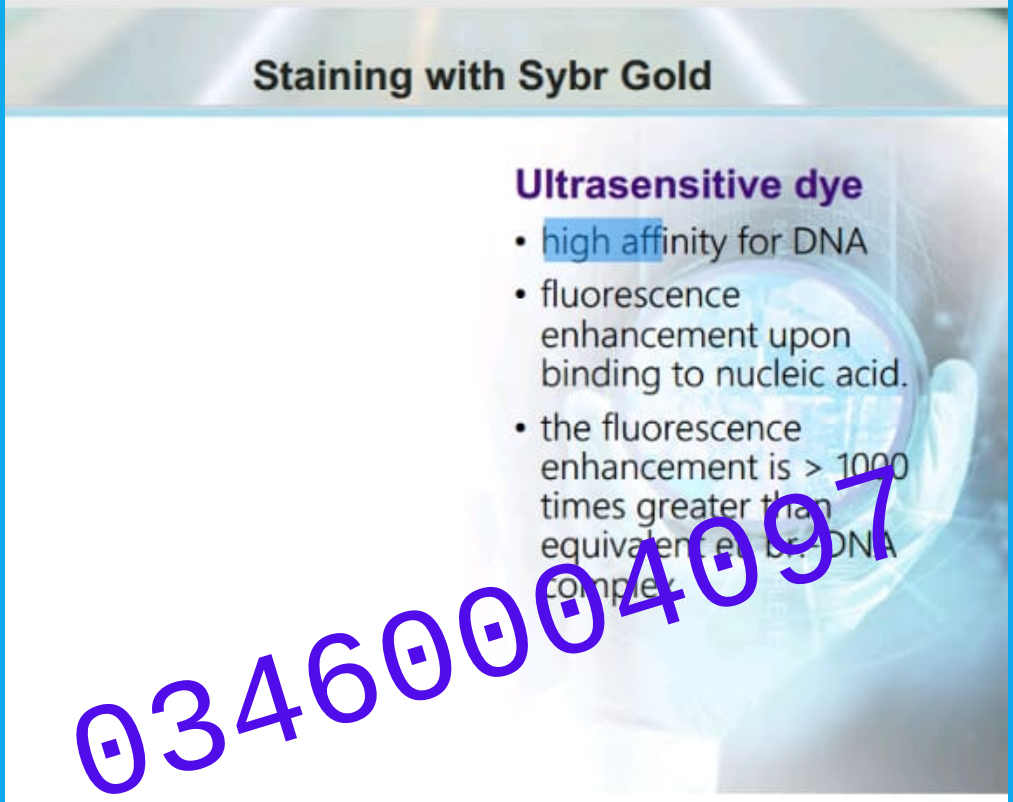
Select the correct option

- ○ RNA
- ● DNA
- ○ Proteins
- ○ Both DNA & RNA

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Click to Save Answer & Move to Next Question

Staining with Sybr Gold



Ultrasensitive dye

- high affinity for DNA
- fluorescence enhancement upon binding to nucleic acid.
- the fluorescence enhancement is > 1000 times greater than equivalent et Br- DNA complex

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Staining with Sybr Gold

Detection limit

- Can detect <20 pg of ds DNA in an agarose gel

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Time Left 64 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 4 of 10 (Start time: 10:27:52 PM, 30 June 2022)

Total Marks: 1

Which of the following fixative is used in routine?

Select the correct option

- Alcohol
- All of the above
- Glutaraldehyde
- Formalin

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Click to Save Answer & Move to Next Question

Method (Commonly used fixatives)

- **Commonly used fixatives are alcohol, formalin, glutaraldehyde, etc.**
- Formalin –routine
- Glutaraldehyde – electron microscopy
- Picric acid(Bouin's solution) – renal & testicular tissue
- Alcohol (Carnoy's fixative) – cytologic smears, endometrial sampling
- Osmium tetraoxide – CNS tissues & electron microscopy

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COMMONLY USED FIXATIVES

- **Factor affecting fixation are temperature, change in pH, penetration of the fixative,**

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Time Left 15 sec(s)

ZOO101:Quiz 2

Quiz Start Time: 10:23 PM, 30 June 2022

Question # 3 of 10 (Start time: 10:25:29 PM, 30 June 2022)

Total Marks: 1

Electrophoresis buffers are prepared and then filled in

Select the correct option

- ○ Glass plate
- ○ DNA sample
- ● Electrophoresis tank
- ○ Combs

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Click to Save Answer & Move to Next Question

Procedure

Electrophoresis buffer

- Prepare sufficient electrophoresis buffer (usually 1x TAE or 0.5x TBE) to fill the electrophoresis tank and to cast the gel.

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Procedure

Agarose Gel preparation

- Prepare a solution of

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BC200203609: MUHAMMAD IQBAL
Time Left 54 sec(s)
ZOO101:Quiz 2
Quiz Start Time: 10:23 PM, 30 June 2022

Question # 2 of 10 (Start time: 10:24:26 PM, 30 June 2022) Total Marks: 1

The resolution of which DNA is better in TAE than in TBE

Select the correct option

- (●) Supercoiled
- ( ) Helical
- ( ) Linear
- ( ) Single stranded

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Click to Save Answer & Move to Next Question

Electrophoresis Buffers

Difference in the results of buffers

- Gels for southern blots are prepared and run with TAE as the electrophoresis buffer.
- The res-olution of supercoiled DNAs is better in TAE than in TBE.

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Electrophoresis Buffers

BUFFER	WORKING SOLUTION	STOCK SOLUTION/LITER
TAE	1x 40 mM Tris-acetate 1mM EDTA	50x 242 g of Tris base 57.1 ml of glacial acetic acid 100 ml of0.5 M EDTA (pH 8.0)
TPE	1x 90mM Tris-phosphate 2mM EDTA	10X 108 g of Tris base [illegible]phoric acid (85%, 1.679 g/ml) 40 ml of 0.5 M EDTA (pH 8.0)

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10:24 4G 30%
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BC200203609: MUHAMMAD IQBAL
Time Left 70 sec(s)
ZOO101:Quiz 2
Quiz Start Time: 10:23 PM, 30 June 2022

Question # 1 of 10 (Start time: 10:23:53 PM, 30 June 2022) Total Marks: 1

Which type of buffers are slightly more expensive than TAE?

Select the correct option

- (●) Both a & b
- ( ) None
- ( ) TPE
- ( ) TBE

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Click to Save Answer & Move to Next Question

through TBE or TPE

Electrophoresis Buffers

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TBE & TPE

- The resolving power of TAE is slightly better than TBE or TPE for high-molecular-weight DNAs and worse for low-molecular-weight DNAs.



Z00101:Quiz 2

Quiz Start Time: 03:01 PM, 30 June 2022

Question # 9 of 10 (Start time: 03:12:03 PM, 30 June 2022)

Total Marks: 1

The pH of Trisphosphate (TPE) buffer at a concentration of 50Mm is

Select the correct option

- 7.5-7.8
- 8.7-8.
- 7.2-7.5
- 5.7-7.8

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Z00101:Quiz 2

Quiz Start Time: 03:01 PM, 30 June 2022